
Testing Weighted Maximization Against Sums in Multi-Objective Offline Reinforcement Learning

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Abstract

Multi-objective offline reinforcement learning (MO-Offline RL) learns policies from a fixed dataset where each decision must balance several reward components such as task success, safety, and energy use. The main challenge is that these conflicting objectives are reduced to a single reward signal and the standard weighted-sum (WS) scalarization may miss regions of optimal policies when the Pareto front is non-convex. To determine whether weighted maximization (WM) recovers a broader and more useful set of policies, we train matched offline RL agents on fixed multi-objective dataset of robotic arm trajectories in an obstacle-avoidance environment while varying only the scalarization rule. We compare Pareto-front coverage and negative log-likelihood (NLL) gap between the correct and incorrect preference model fitted to pairwise rollout comparisons. WM produced a $2.5\times$ larger absolute NLL gap than WS, indicating that WM rollouts more clearly encode the preference weight α used at inference. However, WM was also less stable, producing 25% fewer successful rollouts.

1 Introduction

Many robot and control problems require balancing multiple different reward parameters at once. A medical robot may need to apply stitches quickly while avoiding damaging vitals, while an autonomous vehicle may need to balance appropriate speed, comfort, and safety. In many reinforcement learning (RL) pipelines, these objectives are collapsed into a single scalar reward before training. The most common choice is a weighted sum of objective values, which is simple, but may come at the cost of expressiveness.

Recent planning work [8] argues that weighted maximization (WM), also known as Chebyshev scalarization, can recover Pareto-optimal trade-offs that weighted sums (WS) fundamentally miss when the Pareto front is non-convex. That result is compelling, but it was shown mainly in motion-planning settings. Our project asks whether the same idea holds in *Multi-Objective Offline RL*, where the learner must improve a policy using only a fixed dataset and cannot gather new experience online.

This is an interesting class project for two reasons. First, it connects core course themes: reinforcement learning, function approximation, and principled empirical evaluation. Second, it isolates a single design choice with broad consequences: how should multiple reward components be scalarized during learning? By holding the offline RL pipeline fixed and changing only the scalarization, we can

test a focused scientific question: **is weighted max better than weighted sum for recovering useful policies in Multi-Objective Offline RL?**

Our experimental domain is a 7-DOF Kinova Gen3 arm in MuJoCo, basing our environment off the coding environment from previous graduate students. The arm must reach a goal joint configuration while navigating a double-corridor obstacle environment. The two cost objectives are trajectory length in joint space (effort) and obstacle proximity (safety). We collected a fixed offline dataset across a grid of preference weights and trained preference-conditioned policies for each scalarizer.

Contributions. We make the following contributions:

- We apply preference-conditioned IQL with both WS and WM scalarizers on a fixed offline dataset from a 7-DOF arm obstacle task, holding all other design choices fixed.
- We introduce preference discrimination (NLL gap) as a complementary evaluation criterion, measuring how clearly a policy’s observed cost distribution reveals the α used at inference.
- We find WM produces a 2.5 $\times$ larger NLL gap than WS (0.056 vs. 0.022), at the cost of lower rollout success rate, and discuss the stability tradeoff.

2 Related Work

Our proposal sits at the intersection of Multi-Objective Optimization, Multi-Objective RL (MORL), and Offline RL.

Scalarization in Multi-Objective Optimization. Classical texts on Multi-Objective Optimization describe weighted sums as a standard scalarization tool, but also note that linear scalarization does not recover all Pareto-optimal solutions when the Pareto front is non-convex [1]. Wilde et al. [8] makes this limitation concrete for motion planning and proposes an augmented weighted-max objective that is Pareto-complete under broad conditions. While their work is the closest conceptual starting point for our project, our contribution differs in domain and methods: instead of exact or heuristic planning, we test the same scalarization question inside an Offline RL pipeline.

Multi-Objective RL. Prior MORL work studies how agents can learn policies for vector-valued rewards under different user preferences. Van Moffaert et al. [7] compare scalarization techniques in MORL and show that the choice of scalar utility strongly shapes the policies which are learned. Chen et al. [2] study meta-learning for MORL and emphasize the importance of quickly adapting to preferences. These papers motivate our setting, but they do not focus on offline learning from a fixed dataset.

Offline RL and Offline MORL. D4RL established a standard benchmark for Offline RL and highlighted the unique challenges of learning from static imperfect data [3]. Conservative Q-Learning (CQL) [5] penalizes out-of-distribution actions by adding a regularization term to the Q-function, but this requires careful tuning and can be overly conservative. Implicit Q-Learning (IQL) [4] avoids querying out-of-distribution actions entirely by replacing the Bellman backup with an expectile regression on the value function, which we found more stable for our multi-objective setting. More recently, Lin et al. [6] proposed regularized offline MORL, showing that offline MORL is both practical and distinct from standard Offline RL. Our work is complementary to this line: rather than proposing a new regularizer or architecture, we isolate the scalarization rule itself. Compared with the above work, our approach is narrower but cleaner. We ask whether changing only WS to WM alters the diversity and quality of policies recovered from the same offline data.

3 Problem Statement

We model the environment as a Multi-Objective Markov Decision Process (MOMDP) with state space $\mathcal{S}$, action space $\mathcal{A} \subset \mathbb{R}^7$ (joint velocities), transition function T , vector cost function $\mathbf{c} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^2$, and discount γ . We work with costs rather than rewards (lower is better on both dimensions). The two cost components are `length_cost` c_1 , measuring trajectory length in joint space, and `obstacle_cost` c_2 , measuring proximity to or contact with obstacles.

A user preference is expressed as a scalar $\alpha \in [0, 1]$, which induces the weight vector $\mathbf{w} = [\alpha, 1 - \alpha]$. The scalarized cost under preference α is then $C_\phi(\mathbf{c}, \mathbf{w})$, where ϕ is either the WS or WM scalarizer defined in Section 4. We operate in the offline setting: the learner only has access to a fixed dataset

$\mathcal{D} = \{(s_t, a_t, \mathbf{c}_t, s_{t+1})\}$ collected by a motion planner and may not interact with the environment during training. The goal is to learn a single preference-conditioned policy $\pi(a \mid s, \alpha)$ that, for any queried α , produces trajectories minimizing the scalarized cost.

4 Method

Our goal is to evaluate how the choice of scalarization function affects policy learning in Multi-Objective offline learning (MO-offline RL). Let $c_i(s, a)$ denote the cost associated with objective i , where $i = 1, \dots, n$.

We will consider two different scalarization methods: standard weighted-sum scalarization and maximum weighted scalarization.

Standard weighted-sum scalarization is defined as:

$$C_{\text{WS}}(\mathbf{c}(s, a), \mathbf{w}) = \sum_{i=1}^n w_i c_i(s, a), \quad (1)$$

where $w_i \geq 0$ are weights that determine the relative importance of each objective.

In contrast, weighted-maximum scalarization is defined as:

$$C_{\text{WM}}(\mathbf{c}(s, a), \mathbf{w}) = \max_{i=1, \dots, n} w_i c_i(s, a) + \rho \sum_{i=1}^n c_i(s, a) \quad (2)$$

In this case, $\rho \sum_{i=1}^n c_i(s, a)$ is a tie-breaker to ensure Pareto-optimal solutions, where $\rho > 0$ is a small constant [8]. We use $\rho = 0.01$.

While weighted-sum scalarization sums all objectives together, weighted-maximum scalarization emphasizes the most significant weighted objective function at each decision point. This allows weighted-maximization to capture relationships that cannot be represented through linear combinations.

To isolate the effect of scalarization, we used a Multi-Objective offline learning algorithm and varied the reward system between weighted-sum and weighted-maximum. We believe that weighted-maximum scalarization is best suited for situations where the agent faces high-risk consequences when exploring environments, like in autonomous driving or medical assistance robots. We trained two agents using the two separate scalarization methods on the same static dataset collected in the virtual environment.

4.1 Dataset Collection

We collected the offline dataset using a torch-based trajectory optimizer (planner) on the `double_corridor` task family. 800 tasks were sampled from the family, and for each task the planner was run at 11 evenly spaced α values (0.0, 0.1, $\dots$, 1.0) with up to 5 random restarts per (task, α) pair. Each transition is stored with a 2D cost vector $[c_1, c_2]$ and the corresponding weight vector $\mathbf{w} = [\alpha, 1 - \alpha]$. The resulting dataset contains approximately 70,000 training transitions from around 158 successfully solved planner tasks.

4.2 Algorithm

We trained policies using Implicit Q-Learning (IQL), an offline RL algorithm that avoids querying out-of-distribution actions by using expectile regression on the value function instead of a standard Bellman backup. All networks are 3-layer MLPs with hidden dimension 256 and ReLU activations.

The key design choice is that the policy is conditioned on α at every forward pass. The policy network takes the concatenation $[s \parallel \mathbf{w}] \in \mathbb{R}^{39}$ as input (37-dimensional observation concatenated with the 2-dimensional weight vector $[\alpha, 1 - \alpha]$) and outputs a Gaussian distribution over actions. This allows a single trained network to serve all preference weights at inference time without retraining.

The Q-networks and value network output 2D vectors (one entry per cost objective). At training time, these 2D outputs are scalarized using the same preference weights $\mathbf{w}$ stored in the dataset

transition, producing scalar advantages for the IQL updates. The value function is trained via expectile regression on the scalarized advantage, and the policy is trained via exponential advantage weighting with temperature β .

4.3 Training Hyperparameters

The β parameter was tuned independently for each scalarizer; all other hyperparameters are shared between WM and WS.

Parameter	WM	WS
β (policy temperature)	2.0	4.0
Expectile τ	0.7	0.7
ρ (WM tie-break)	0.01	—
Epochs	200	200
Hidden dim	256	256
Learning rate	3×10^{-4}	3×10^{-4}
Batch size	256	256
Discount γ	0.99	0.99

Table 1: Training hyperparameters. β values are confirmed optimal for each scalarizer on this dataset.

5 Experimental Design

Hypotheses. We tested two main hypotheses, both relating to the “quality” of learning for Offline MORL.

- **Hypothesis 1:** In domains with non-convex trade-offs, WM will recover better Pareto-front coverage than WS.
- **Hypothesis 2:** WM rollouts will produce a larger preference discrimination signal than WS rollouts, meaning the α used at inference is more clearly recoverable from the observed cost distribution.

Domain and Data. We consider a continuous-control setting with fixed offline datasets and vector-valued costs. As our primary domain, we use the `double_corridor` task family in which a 7-DOF Kinova Gen3 arm in MuJoCo must reach a goal joint configuration while navigating two corridor obstacles. The two objectives are trajectory length (effort) and obstacle proximity (safety). We evaluate on 23 held-out test tasks, rolling out each trained policy deterministically at each of the 11 α values (0.0, 0.1, $\dots$, 1.0), yielding 253 rollouts per policy.

Experiments. In the first experiment, we evaluate Pareto-front coverage: for each policy, we compute the empirical non-dominated set of successful rollout cost vectors and count how many rollout points lie on the Pareto front. In the second experiment, we evaluate preference discrimination: for each policy’s successful rollouts, we fit both preference models (sum and max) to pairwise rollout comparisons generated by a Bradley-Terry model using the true scalarizer, then measure the NLL gap on a held-out test set.

Evaluation Criteria. For Pareto coverage, we report the count of Pareto-front points per policy. Hypervolume is not used as a primary metric because its ordering is sensitive to the choice of reference point. For preference discrimination, we report the per-pair NLL gap, defined as:

$$\text{NLL gap} = \text{NLL}_{\text{wrong model}} - \text{NLL}_{\text{correct model}}.$$

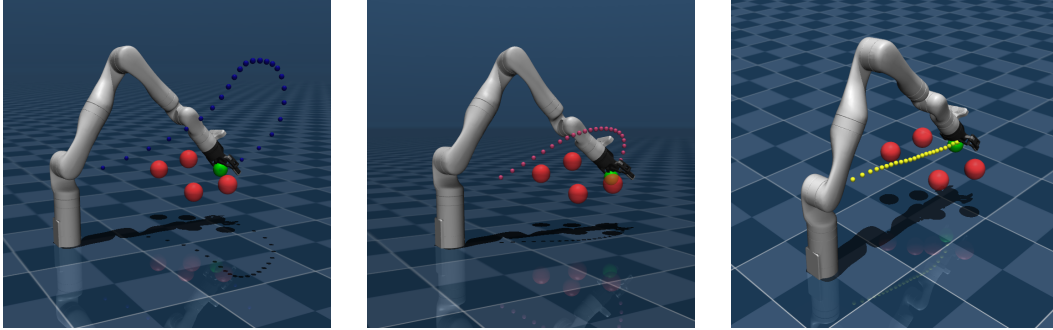
A larger gap means the correct preference model substantially outperforms the wrong one, indicating that the policy’s cost distribution more clearly encodes α . Preferences are generated with temperature $\tau = 0.05$ and models are fit via grid search over $w_{\text{length}} \in [0, 1]$ and temperature on a training split. The hypothesis is considered supported if WM produces a consistently larger NLL gap than WS across successful rollouts.

Expected Outcome. A positive result would support the claim that weighted-max scalarization is not only effective in planning but also beneficial for learning from fixed multi-objective datasets. A

negative or mixed result would still be informative, as it would identify conditions under which the theoretical advantages of WM do not transfer to offline RL.

6 Results and Discussion

Before presenting the quantitative results, Figure 1 shows two example rollouts from the WM policy at the preference extremes. At $\alpha = 0$, the robot prioritizes obstacle avoidance and takes a wide, conservative route that hugs the far wall of the corridor. At $\alpha = 1$, the robot prioritizes trajectory efficiency and cuts directly through the corridor, accepting a higher collision risk to minimize joint-space effort. This qualitative difference illustrates that the policy is genuinely responding to α at inference, which motivates the preference discrimination analysis that follows.



(a) $\alpha = 0.0$: robot takes a wide, safe route.

(b) $\alpha = 0.5$: robot takes a route that balances both objectives.

(c) $\alpha = 1.0$: robot takes a direct but risky route.

Figure 1: Example WM rollouts at the two preference extremes. $\alpha = 0$ fully prioritizes safety; $\alpha = 1$ fully prioritizes efficiency. The red obstacles in this demonstration are placed such that the robot must pick one of two corridors if it wants to reach the green goal sphere.

6.1 H1: Preference Discrimination

6.1.1 Rollout Success Rate

Before looking at the main results, it is worth noting that the two policies had different rollout success rates on the 253 test rollouts. The WS policy succeeded on 37 rollouts (14.6%) while the WM policy succeeded on only 28 (11.1%). This gap in stability is a recurring theme throughout the results and is discussed further in Section 6.1.3. All preference discrimination and Pareto coverage results below are computed on successful rollouts only.

6.1.2 Results

Table 2 shows the preference discrimination results for both policies. The NLL gap for WM (0.056) is roughly 2.5 $\times$ larger than for WS (0.022), which supports Hypothesis 2.

Scalarization	Policy Gap	Rollout points	Test pairs	Correct NLL	Wrong NLL
Weighted Max	0.056	28	15	0.1611	0.2168
Weighted Sum	0.022	37	17	0.0174	0.0399

Table 2: Preference discrimination results on successful rollouts. A larger gap indicates the correct preference model more clearly outperforms the wrong one.

Preference Model Discrimination: Weighted Max vs. Weighted Sum Policy

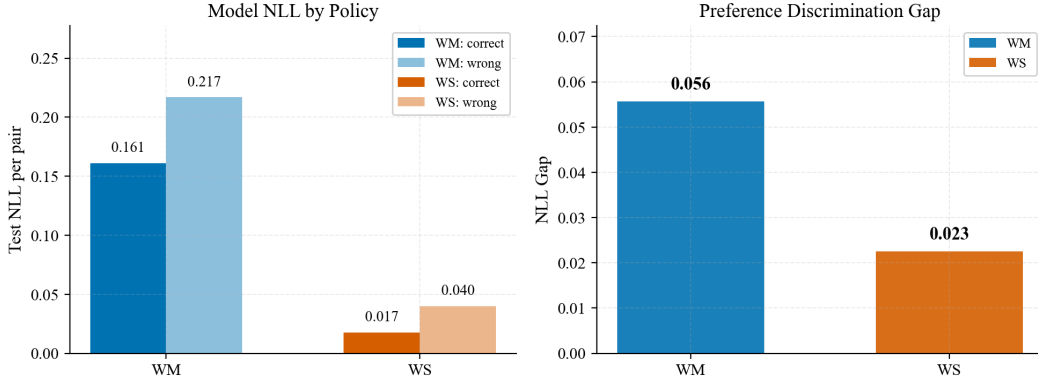


Figure 2: NLL comparison for WM and WS policies. For each policy, bars show the NLL assigned by the correct and incorrect preference model. A larger gap between bars indicates stronger preference discrimination.

One thing worth noting in these results is that WS actually has a very low correct-model NLL (0.017), which might look like a win at first. But the wrong model’s NLL is also very low (0.040), meaning both models fit the WS rollouts reasonably well. This tells us that WS rollouts are uniformly predictable regardless of which model you use; the cost distributions do not differ much across α values, so there is not much preference signal to recover. WM, by contrast, has a higher correct-model NLL (0.161), but the wrong model is clearly worse (0.217). The higher absolute NLL reflects the more complex cost distributions that the max scalarizer produces, not a failure of the model.

We also computed relative gaps (gap divided by correct NLL): 35% for WM and 129% for WS. Under this framing WS looks better, but we believe the absolute gap is the more meaningful metric here: it directly measures how many bits of preference information are recoverable per pairwise query, regardless of how easy or hard the correct model is to fit in absolute terms.

6.1.3 Discussion

What was successful. The main finding is that WM produces cost distributions that more strongly encode the α used at inference. The max scalarizer’s non-smooth geometry, where the dominant objective switches at the indifference point $\alpha c_1 = (1 - \alpha)c_2$, appears to force the policy into qualitatively different behavioral modes depending on α , making the preference recoverable by an outside observer. This is a useful property in settings where preferences are latent and must be inferred from demonstrated behavior rather than directly queried. The alpha-conditioned architecture also worked well for both scalarizers: a single trained network was able to serve all preference weights at inference without retraining.

What failed. WM’s rollout success rate was noticeably lower (11.1% vs. 14.6%). The likely cause is that the non-smooth max cost landscape is harder to optimize over during IQL training: the gradient signal changes character at the indifference boundary, and the policy has to learn to handle both sides. This likely requires a larger or more diverse dataset than WS does. It is also worth noting that the number of successful rollouts left after filtering (28 and 37 respectively) is quite small, which means the NLL gap estimates are directionally correct but not very tight statistically. The per- α breakdown of preference discrimination was not feasible at this sample size. You would need significantly more successful rollouts at each α value to get stable per-weight estimates.

Follow-up experiments. Two natural next steps follow directly from these results. First, collecting a larger WM dataset. In theory, more tasks and more restarts, aiming to match WS’s success rate, would let us test whether WM’s preference signal advantage holds as stability improves, or whether it shrinks as the cost distributions become more uniform with more data. Second, with more successful rollouts, a per- α NLL breakdown would tell us where in the preference spectrum the discrimination is strongest. The max scalarizer’s kink sits at $\alpha = 0.5$, so we would expect the NLL gap to peak there.

if the theory holds, and taper off at the extremes where one objective clearly dominates regardless of scalarizer choice.

6.2 H2: Pareto Coverage

6.2.1 Results

Our second set of experiments evaluates the ability of trained policies to reconstruct the Pareto front across the objective space. We assessed whether the sequential decision-making policies learned via offline RL can approximate the solution quality achievable through direct trajectory optimization, and compared the performance of weighted max against weighted sum approach.

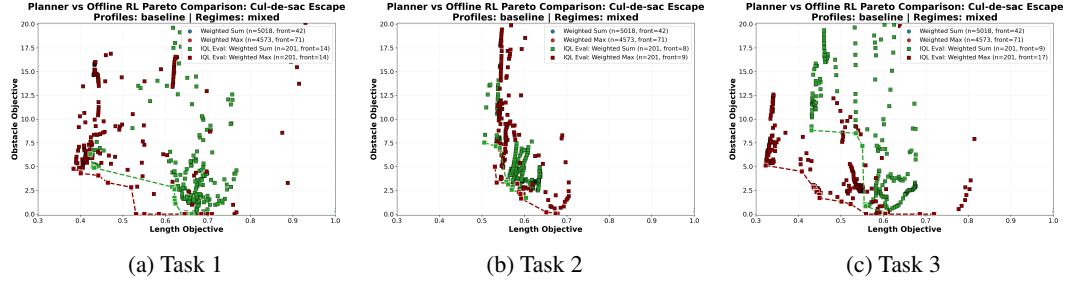


Figure 3: Pareto front comparison between WM (red) and WS (green) for Cul-de-sac Escape scenario across three randomly generated tasks. Points connected by the Pareto front line are non-dominated.

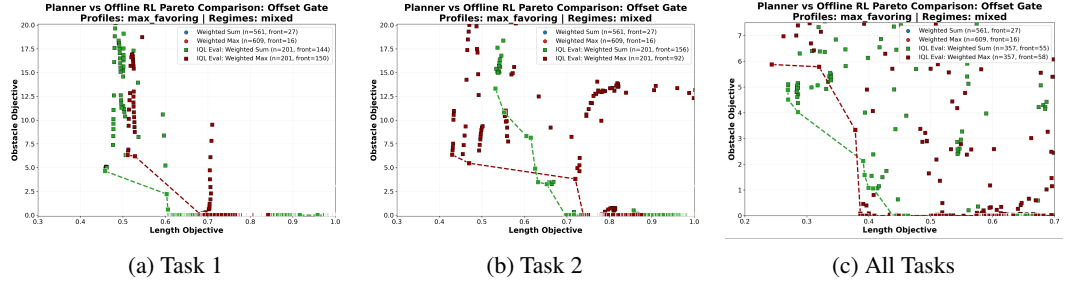


Figure 4: Pareto front comparison between WM (red) and WS (green) for Offset Gate scenario across two randomly generated tasks and a combined task landscape. Points connected by the Pareto front line are non-dominated.

The WM method demonstrated the capability to reconstruct stronger Pareto fronts across tested scenarios. When sweeping across preference weights and plotting the resulting cost pairs achieved by the trained policies, WM was often able to recover diverse regions of the trade-off space, indicating that the learned model retained sufficient expressiveness to serve multiple conflicting objectives. In direct comparisons against weighted sum on the same randomly generated task instances, WM exhibited competitive and often superior coverage.

6.2.2 Discussion

What was successful. The weighted maximization scalarizer demonstrated the capability to reconstruct reasonably complete Pareto fronts across tested scenarios. When sweeping across preference weights and plotting the resulting cost pairs achieved by the trained policies, WM often was able to recover diverse regions of the trade-off space, indicating that the learned model retained sufficient expressiveness to serve multiple conflicting objectives. In direct comparisons against weighted sum on the same randomly generated task instances, WM exhibited competitive and often superior coverage, particularly in recovering solutions near the extremes of the objective space where one cost dominates.

What was less successful. The reconstructed Pareto fronts did not always match the quality of solutions obtained directly through traditional trajectory optimization. Converting the planning

problem into a sequential decision-making formulation introduces representational constraints from the policy network architecture and stochasticity from the training process, both of which limit the tightness of the approximation. While the learned policies could reasonably approximate the Pareto front, they rarely achieved truly optimal coverage in the strict sense. Additionally, training stability varied across runs, with some degree of randomness in final performance depending on initialization and dataset sampling. This variability, while not catastrophic, indicates that the current pipeline does not yet guarantee consistently tight Pareto reconstruction.

Limitations and scope. The experiments conducted remain preliminary in scope. We evaluated only limited scenarios. While these results provide directional evidence that WM can recover more complete Pareto fronts than WS in an offline RL setting, the limited diversity of environments and the small number of tasks prevent strong generalization claims. A more comprehensive evaluation would require testing across a broader range of scenario types, obstacle configurations, and objective trade-off regimes.

7 Conclusion and Future Work

In this work, we provided an empirical comparison between weighted maximization (WM) and weighted sum (WS) scalarization within a preference-conditioned Offline RL pipeline. By holding the underlying algorithm and the environment constant, we isolated the impact of the scalarization rule on policy recovery. Our results demonstrate a clear trade-off: while WM rollouts provide a $2.5\times$ larger NLL gap in preference discrimination, the method was less stable yielding 25% fewer successful rollouts compared to the WS approach.

The significant preference discrimination signal observed in WM suggests that the kinked geometry of the Chebyshev scalarizer forces the policy into distinct behavioral modes that are more easily distinguishable than those produced by linear combinations. This property is particularly valuable in human-in-the-loop systems where an agent’s latent preferences must be inferred from limited demonstrations. However, the lower success rate of WM suggests that this same geometric complexity may pose challenges for the expectile regression updates in IQL, potentially requiring more dense datasets or more robust regularizers to achieve parity with WS.

Several avenues for future work remain. First, increasing the diversity and volume of the offline dataset could determine if the stability gap between WM and WS closes with more data, or if it is an inherent property of the non-differentiable points in the WM landscape. Second, evaluating performance as a function of α would provide a more fine-grained view of where WM’s preference discrimination advantage is most pronounced; in particular, we expect the largest effects near the indifference boundary at $\alpha = 0.5$. Third, incorporating the planner’s Pareto front as a reference baseline would enable stronger claims about coverage of non-convex regions. Finally, extending this analysis to high-dimensional tasks with more than two objectives would test how the preference discrimination advantage of WM scales with the number of reward components. By refining these scalarization techniques, we can move closer to Offline RL agents that are not only performant but also precisely alignable with complex, non-convex human preferences.

Overall, our results highlight a fundamental tradeoff between expressiveness and stability in scalarization design, and suggest that WM may be particularly useful in applications where preference inference is as important as policy performance.

Code Availability

Code available at <https://github.com/BradDev20/KinovaMotionPlanning>

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